

QP

YEAR 9 Science

Common Assessment 1

September 2026

Working Scientifically

B1 Genes and Inheritance

C1 Periodic Table (Atomic Structure only)

45 Minutes

45 Marks

Name	
Class	
Teacher	

INSTRUCTIONS TO CANDIDATES

Write your name, class and teacher's name in the box above.

Do not open this examination paper until instructed to do so.

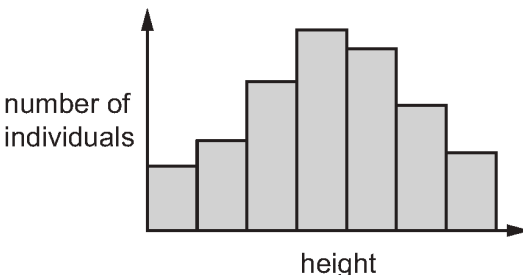
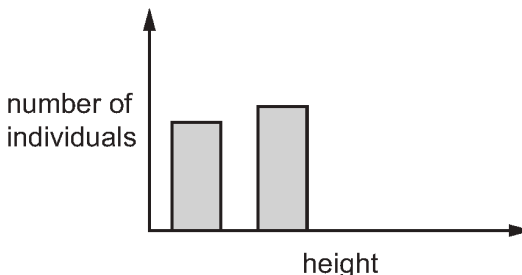
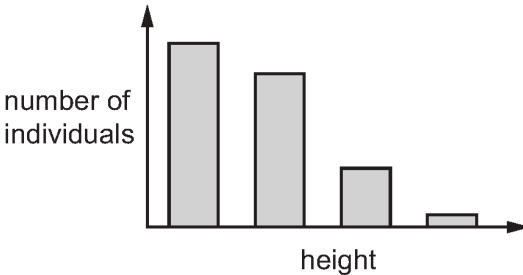
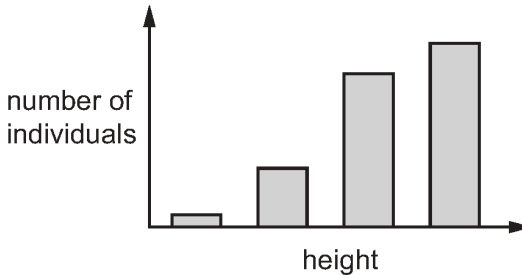
Write your answers for the multiple choice questions on page 5.

MCQ Score	/10
Structured Questions Score	/35
Total	/45
Percentage	%

This document has **16** pages.

[Turn over]

Section A: Multiple Choice Questions - 10 marks

1.	Which graph represents variation in the height of humans? A  B  C  D 
2.	Which line shows the structures in increasing size? A chromosome → gamete → gene → nucleus B chromosome → nucleus → gene → gamete C gene → chromosome → gamete → nucleus D gene → chromosome → nucleus → gamete
3.	Which statement about human gametes is correct? A All egg cells contain an X chromosome. B All sperm cells contain a Y chromosome. C All egg cells contain X and Y chromosomes. D All sperm cells contain X and Y chromosomes.

4.	What is the correct term for an allele that is always expressed in the phenotype if it is present? A dominant B heterozygous C homozygous D recessive
5.	A monohybrid cross between tall (TT) and short (tt) pea plants produces 100% tall (Tt) offspring. What do the letters T and t represent? A alleles B chromosomes C genes D genotypes
6.	Two parents with genotypes of BB and Bb are crossed. What is the predicted ratio of genotypes in the offspring? A 1 BB : 1 Bb B 1 Bb : 1 bb C 3 BB : 1 Bb D 3 Bb : 1 BB

7.

In a plant, the allele for red flowers is dominant to the allele for yellow flowers. A heterozygous red-flowered plant is crossed with a homozygous yellow-flowered plant.

Which statement about the offspring is correct?

- A 25% will have red flowers and 75% will have yellow flowers.
- B 50% will have red flowers and 50% will have yellow flowers.
- C 75% will have red flowers and 25% will have yellow flowers.
- D 100% will have red flowers and 0% will have yellow flowers.

8.

A farmer wants sheep with a high yield of fleece that will give fine-quality wool.

The table shows six of the sheep available for breeding.

sheep number	sex of sheep	fleece yield	wool quality
1	female	medium	fine
2	female	medium	coarse
3	female	high	fine
4	male	low	fine
5	male	low	coarse
6	male	medium	fine

Which sheep should the farmer select to breed together?

- A 1 and 3
- B 1 and 4
- C 3 and 4
- D 3 and 6

9.	Atoms consist of protons, neutrons and electrons. Which statement is correct? A An electron has a relative mass of 1 and a negative charge. B A proton has a relative mass of 1 and a positive charge. C A proton has a relative mass of less than 1 and a negative charge. D A neutron has a relative mass of less than 1 and no charge.
10.	What do elements in the same period in the Periodic Table have in common? A number of electron shells B number of electrons in the outer shell C number of neutrons in the nucleus D proton number

Please write your multiple choice answer (A,B C or D) in the table below.

1		6	
2		7	
3		8	
4		9	
5		10	

Section B: Structured Questions - 35 marks

Question 1

The photographs below show pupils investigating the movement of objects on ramps.



Plan an investigation into the factors affecting the movement of objects on ramps.

You can use any objects and any surfaces you like, and any other equipment you need.

In the box below, write a short draft of **one** question you could plan to investigate about the movement of objects on ramps.

Use your draft to help you answer the following questions.

- (a) Give **one** factor you could change as you carry out your investigation (the independent variable).

.....
.....

1 mark

- (b) What factor would you observe or measure to collect your results (the dependent variable) and what equipment would you use to measure them?

The factor I would observe or measure is

.....

1 mark

The measuring equipment I would use is

.....

1 mark

- (c) Give **one** factor you should keep the same to make your test fair.

.....

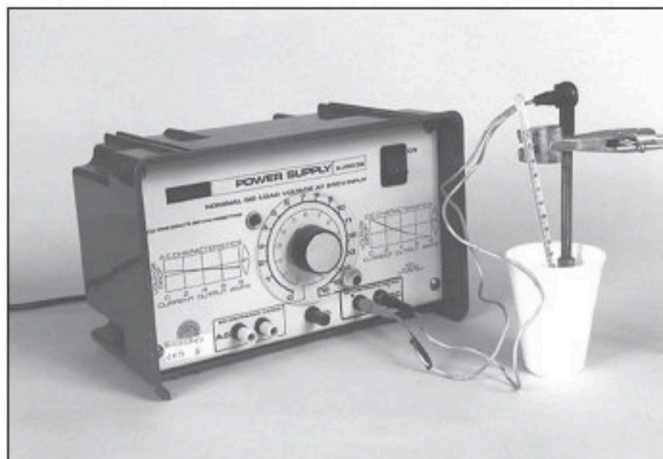
.....

1 mark

[Total: /4]

Question 2

John used an electrical heater to heat a cup of water. He kept stirring the water. When the temperature reached 20°C, he started his stopwatch and measured the temperature of the water every half minute.

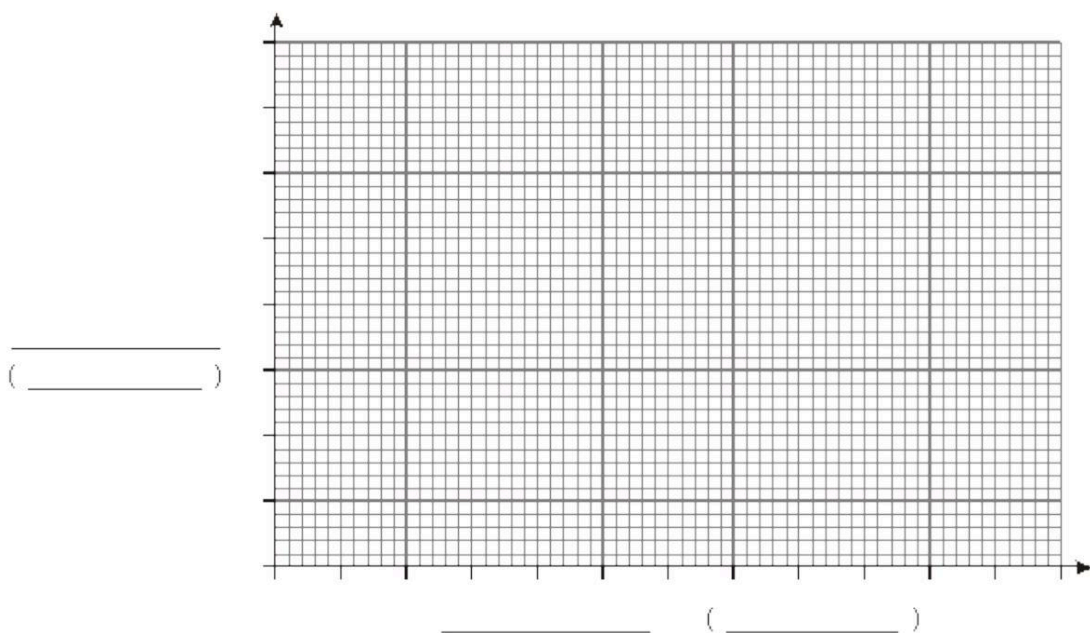


He switched off the heater after 4 minutes, but continued to record the temperature. His results are shown in the table.

One measurement is missing and another appears to be wrong.

Time (minutes)	Temperature (°C)
0.0	20
0.5	26
1.0	31
1.5	36
2.0	41
2.5	46
3.0	
3.5	57
4.0	56
4.5	58
5.0	59
5.5	59

- (a) Use the results in the table to draw a graph on the grid.
Label the axes.
Plot the points and draw a smooth curve of best fit.



4 marks

- (b) From your curve, estimate the temperature of the water after **three** minutes.

.....°C

1 mark

[Total: /5]

Question 3

(a) Fig. 13.1 is a drawing of the chromosomes in a human.

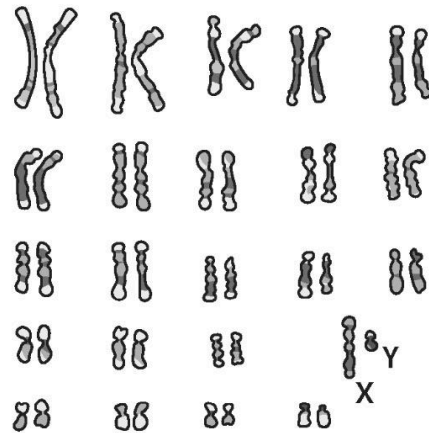


Fig. 13.1

(i) State the number of chromosomes shown in Fig. 13.1.

.....

[1]

(ii) State whether the drawing of the chromosomes shown in Fig. 13.1 is from a male or female.

Explain your answer.

.....

..... [1]

(iii) Describe how the sex of the person in your answer in (a)(ii) is inherited.

.....

.....

..... [2]

(b) Complete the definition of the term *chromosome* using the words from the list.

You may use each word once, more than once or not at all.

DNA	embryos	gametes
genes	nuclei	mitochondria

A chromosome is a thread of , made up of a string
of

[2]

(c) The transmission of genetic material occurs during fertilisation.

Describe the process of fertilisation.

.....
.....
..... [2]

[Total: /8]

Question 4

(a)

Alleles are made of DNA.

(i) Define the term allele.

..... [1]

(ii) State where alleles are found in a cell.

..... [1]

- (b) A class investigated the number of students that are able to roll their tongues.

Fig. 10.1 is a graph of the results.

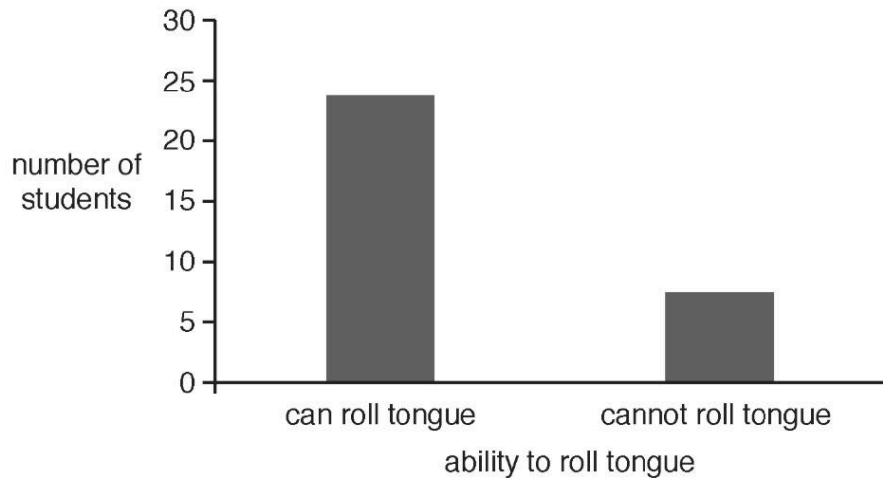


Fig. 10.1

Use evidence from Fig. 10.1 to explain why this is an example of **discontinuous** variation.

.....
..... [1]

- (c) The list shows some examples of different types of variation.

Place a tick (✓) in the boxes to show **all** the examples of **continuous** variation.

height	
foot length	
sex (gender)	
types of teeth	
mass	

[2]

- (d) Fig. 10.2 is a photograph of a giraffe, a mammal. Giraffes eat the leaves from branches on trees.



Fig. 10.2

The ancestors of giraffes had shorter necks.

Describe how giraffes developed long necks by natural selection.

Use the words **variation**, **competition** and **alleles** in your answer.

.....

.....

.....

.....

.....

..... [4]

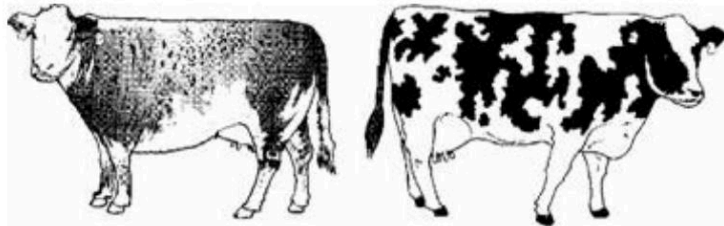
[Total: /9]

Question 5

Herefords and Friesians are two breeds of cattle.

Herefords produce high quality meat. Friesians produce lots of milk.

The drawings below show a Hereford cow and a Friesian cow.



Hereford cow

Friesian cow

- (a) (i) The two breeds of cattle are different in appearance from each other. What causes the variation between the two breeds of cattle?

.....
.....

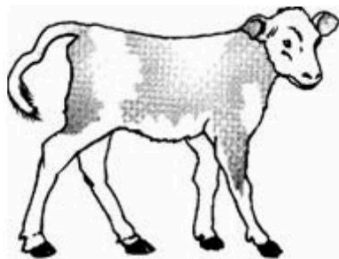
1 mark

- (ii) Suggest **two** environmental factors which can affect the amount of meat or milk cattle produce.

1.
.....
2.
.....

2 marks

- (b) The drawing shows a calf produced by mating a Hereford bull with a Friesian cow. Cattle bred in this way will produce both high quality meat and a high milk yield.



- (i) What term is used to describe this deliberate mating of two different breeds of animals to produce offspring with particular characteristics?

.....

1 mark

- (ii) Farmers want their cattle to produce high quality meat and a high milk yield. Suggest **one other** characteristic which farmers might want their cattle to have.

.....

.....

1 mark

[Total: /5]

Question 6

- (a) Fig. 11.1 shows part of Group I of the Periodic Table.

3 Li lithium 7
11 Na sodium 23
19 K potassium 39
37 Rb rubidium 85

Fig. 11.1

(i) State the electronic structure of a potassium atom.

..... [1]

(ii) Describe how the electronic structure of potassium is related to its group number.

.....

..... [1]

(b)

Potassium has a proton number of 19.

Complete Fig. 5.2 to show the electronic structure of a potassium atom.

The inner shell electrons have been drawn for you.

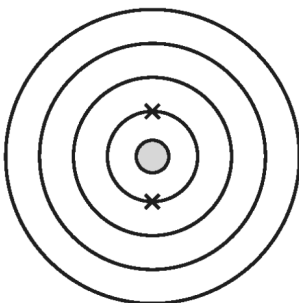


Fig. 5.2

[2]

[Total: /4]

- END OF ASSESSMENT -

The Periodic Table of Elements

57	La	lanthanum	139
58	Ce	cerium	140
59	Pr	praseodymium	141
60	Nd	neodymium	144
61	Pm	promethium	—
62	Sm	samarium	150
63	Eu	euroium	152
64	Gd	gadolinium	157
65	Tb	terbium	159
66	Dy	dysprosium	163
67	Ho	holmium	165
68	Er	erbium	167
69	Tm	thulium	169
70	Yb	yterbium	173
71	Lu	lutetium	175
72	Hf	hafnium	178
73	Ta	tantalum	181
74	W	tungsten	184
75	Re	rhenium	186
76	Os	osmium	190
77	Ir	iridium	192
78	Pt	platinum	195
79	Au	gold	197
80	Hg	mercury	201
81	Tl	thallium	204
82	Pb	lead	207
83	Bi	bismuth	209
84	Po	polonium	—
85	At	astatine	—
86	Rn	radon	—
87	Fr	francium	—
88	Ra	radium	—
89	Ac	actinium	227
90	Th	thorium	232
91	Pa	protactinium	231
92	U	uranium	238
93	Np	neptunium	—
94	Pu	plutonium	—
95	Am	americium	—
96	Cm	curium	—
97	Bk	berkelium	—
98	Cf	californium	—
99	Es	einsteinium	—
100	Fm	fermium	—
101	Md	mendelevium	—
102	No	nobelium	—
103	Lr	lawrencium	—

The volume of one mole of any gas is 24 dm^3 at room temperature and pressure (r.t.p.).

MS

Multiple Choice Questions

1.	A	6.	A
2.	D	7.	B
3.	A	8.	D
4.	A	9.	B
5.	A	10.	A

Structured Questions

1.	(a) any suitable independent variable such as - the surface - the angle of the slope - the kind of object - the size of the push <i>accept specific variations in objects, such as, 'weight' or 'mass' or 'surface area' or 'type of trainer sole' or 'type of shoe'</i> 1 (L4) (b) any suitable dependent variable such as - the distance travelled - the time to move down the ramp - the force needed to start the object moving - the angle of the ramp at which the object starts moving <i>accept 'the time to reach a given point'</i> <i>accept 'angle or height of ramp'</i> <i>accept 'speed'</i> <i>a dependent variable (DV) without an independent variable (IV) can gain credit</i> 1 (L4)
----	---

any appropriate equipment to measure the dependent variable such as

- ruler **or** metre rule
- stopwatch **or** timer **or** light gates
- newton meter
- protractor

accept 'tape measure'

accept 'clock'

*do **not** accept a measurement strategy if a DV is not given
or is incorrect*

1 (L4)

(c) any appropriate control variable such as

- the object used
- the angle of the slope
- the surface used
- the height of the ramp
- the length of the ramp

accept 'distance travelled'

*only give credit for a control variable which does
not conflict with the suggested investigation*

1 (L4)

[4]

2.	(a) • X-axis: time (minutes) Y-axis: temperature ($^{\circ}\text{C}$) 1 (L7) • appropriate scales such as 2 cm represents 10°C, 2 cm represents 1 minute 1 (L7) • 11 points plotted to within half a small square 1 (L7) • a smooth curve of best fit (not dot-to-dot) <i>pupils can gain credit for correct responses to other parts if the axes are wrongly labelled both answers are required for the mark units are required for the mark at least one zero must be included at the origin</i> 1 (L7) (b) an answer consistent with the drawn curve <i>accept a tolerance of one small square</i> 1 (L6) [5]															
3.	<table><tr><td>(a)(i)</td><td>46 ;</td><td>1</td></tr><tr><td>(a)(ii)</td><td>male and the presence of XY chromosomes ;</td><td>1</td></tr><tr><td>(a)(iii)</td><td>X inherited from, mother / egg / female gamete ; Y inherited from, father / sperm / male gamete ;</td><td>2</td></tr><tr><td>(b)</td><td><u>DNA</u> ; <u>genes</u> ;</td><td>2</td></tr><tr><td>(c)</td><td>joining of nuclei ; (of) sperm and egg / male and female gamete / sex cells ;</td><td>2</td></tr></table>	(a)(i)	46 ;	1	(a)(ii)	male and the presence of XY chromosomes ;	1	(a)(iii)	X inherited from, mother / egg / female gamete ; Y inherited from, father / sperm / male gamete ;	2	(b)	<u>DNA</u> ; <u>genes</u> ;	2	(c)	joining of nuclei ; (of) sperm and egg / male and female gamete / sex cells ;	2
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5.

(a) (i) any **one** from

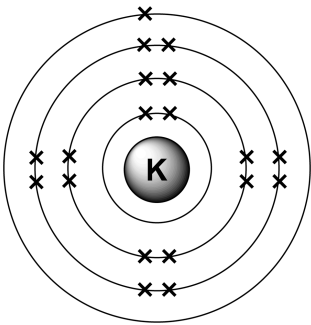
- they have inherited different characteristics
accept 'selective breeding'
- they have different genes
*accept 'they have different DNA **or** alleles'*

1

(ii) any **two** from

- amount of food
accept 'amount of grass'
- type of food
- amount of water
- levels of disease **or** quality of health care
*accept 'disease' **or** 'health care'*
- levels of stress
- temperature
accept 'climate'
*accept 'amount of exercise' **or** 'amount of space'*
accept 'their hormone levels'

2

	(b) (i) selective breeding <i>accept 'artificial selection' or 'cross breeding'</i> <i>do not accept 'inter-breeding'</i> 1 (ii) any one from - resistance to disease <i>do not accept 'good health'</i> - fast growth rate - high fertility <i>accept 'fertile'</i> - long life - good skin for making leather - docile temperament - hardiness or resistance to weather <i>accept 'prize-winning appearance'</i> <i>accept 'females' or 'heifers'</i> <i>accept 'polled' or 'no horns'</i> 1 [5]				
6.	<table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 15%; padding: 5px;">(a)(i)</td> <td style="padding: 5px;">2.8.8.1;</td> </tr> <tr> <td style="padding: 5px;">(a)(ii)</td> <td style="padding: 5px;">number of outer shell electrons is the same as group number;</td> </tr> </table> 19 electrons shown ; (2).8.8.1 ; 	(a)(i)	2.8.8.1;	(a)(ii)	number of outer shell electrons is the same as group number;
(a)(i)	2.8.8.1;				
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